

Satyam Sangeet PhD Candidate (Founder - CompObelisk)

India

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Research Gate

Company Website

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Education

2024/01 – present
Sydney, Australia

Doctor of Philosophy (Chronophysics & Chronobiology)
University of Sydney

2018/07 – 2020/06
Bhopal, India

Master of Technology (Biotechnology)
Maulana Azad National Institute of Technology

- Average GPA - 9.59/10
- Master Dissertation - "**Effect of Stochasticity on neuronal signaling**" under the guidance of Dr. Suhita Nadkarni at Indian Institute of Science Education and Research (IISER) - Pune (12 months)

2014/08 – 2018/05
Pune, India

Bachelor of Technology (Biotechnology)
Dr D.Y. Patil Biotechnology and Bioinformatics Institute

- Average GPA - 9.42/10
- Bachelor Dissertation - "**Development of Microbial zombies using synthesized metal nanoparticles**" under the guidance of Dr. Swapnil Gaikwad at Dr. D.Y. Patil Biotechnology & Bioinformatics Institute (6 months)
- Sports Secretary

Awards

2026

Postgraduate Research Support Scheme (PRSS) Travel Grant
University of Sydney

Awarded the PRSS Travel grant for the European Sleep Research Society (ESRS) in Maastricht, Netherlands

2025

Postgraduate Research Support Scheme (PRSS) Travel Grant
University of Sydney

Awarded the PRSS Travel grant for the World Sleep Conference in Singapore (September 2025)

2024

Faculty of Science Research Scholarship
University of Sydney

Awarded the Faculty of Science Research Scholarship for pursuing PhD in the field of Computational Neuroscience

2024

Centre for Chronic Diseases of Ageing (CCDA)
Woolcock Institute of Medical Research

Awarded the top up scholarship from Woolcock Institute of Medical Research to pursue my PhD

- 2021/01/05 **Har Gobind Khorana IYBA (Innovative Young Biotechnologist Award) Junior Research Fellow Award**
Department of Biotechnology (DBT), Government of India
Awarded the prestigious Har Gobind Khorana IYBA Junior Research Fellow Award for the project funded by the Department of Biotechnology (DBT), Government of India, to be conducted at Indian Institute of Science Education and Research (IISER) - Kolkata
- 2020/10/12 **Junior Research Fellow**
Department of Biotechnology (DBT), Government of India
Awarded the Junior Research Fellow Award for the project funded by the Department of Biotechnology (DBT), Government of India, to be conducted at Indian Institute of Science Education and Research (IISER) - Bhopal
- 2018/07/19 **GATE Scholarship**
AICTE, Government of India
Awarded the GATE scholarship by AICTE, Government of India for the completion of Master studies
- 2018 **Subroto Memorial Scholarship**
Indian Air Force Benevolent Association
Awarded the Subroto Memorial Scholarship for academic excellence in Bachelor Studies by Indian Air Force.
- 2013 **Subroto Memorial Scholarship**
Indian Air Force Benevolent Association
Awarded the Subroto Memorial Scholarship for academic excellence in Middle and High school
- 2011 **Jagdish Bose National Science Talent Search**
Department of Science and Technology (DST), Government of India
Represented my high school at JBNSTS organized by the Department of Science and Technology (DST), Government of India

Independent Technical Projects

EVOLVE

Developed a web-server that allows an easy user interface to perform feature calculation and machine learning model training on viral protein sequences to predict future mutations in variants of the virus. This webserver is developed based on my previous published work: "[*Quantifying Mutational Response to Track the Evolution of SARS-CoV-2 Spike Variants: Introducing a Statistical-Mechanics-Guided Machine Learning Method*](#) [!\[\]\(99f58673407353e96a019fbca558fd72_img.jpg\)](#)"

SIMulation ANALysis (SIMANA)

Developed SIMANA, a webtool **for analyzing and visualizing molecular dynamics simulation data** along with performing multiple drug-discovery related calculations. The platform aims to streamline the analysis process and enable deeper insights into molecular behavior.

ARIAS (Article Research Intelligence And Summarization)

Developed a webserver that allows easy summarization of the research articles. The webserver also allows the user to interact with the research article and ask specific questions based on the published work. I integrated the language model ChatGPT to make the summarization and questions asked by the user to be interactive with the AI.

Research Experience

- 2021/01 – 2023/12 **Tracking Mutational Response of SARS-CoV-2 Variants**
Har Gobind Khorana Junior Research Fellow | Indian Institute of Science Education and Research (Kolkata) | PI: Dr. Susmita Roy
- Planned and organized the experiments to investigate the genomic fluctuations of SARS-CoV-2 sequences
 - Visualized the mutational pattern of different Variant of Concerns (VOCs) and Variant of Interests (VOIs)
 - Developed mathematical parameter and multi-layer perceptron model to quantify the mutational response
- 2020/10 – 2020/12 **Synthesis of Double Knot Toxins (DkTx) and sub-variants for TRPV1 Activation**
Junior Research Fellow | Indian Institute of Science Education and Research (Bhopal) | PI: Dr. Jeet Kalia
- Reviewed pieces of literature on Double knot toxins (DkTx) and their role in ion channel activation and concluded that the development of hybrid variants of DkTx will allow investigating the mechanism of ion channel activation.
 - Purified DkTx and different variants using HPLC and buffer preparations.
 - Handled the culture of *Xenopus laevis*
- 2019/05 – 2020/06 **Effect of stochasticity on Neuronal Signalling**
M.Tech Dissertation | Indian Institute of Science Education and Research (Pune) | PI: Dr. Suhita Nadkarni
- Constructed the neural signaling model of CA3-CA1 hippocampal synapse.
 - Inspected the effect of noise on neural signaling by varying ion channel numbers.
- 2019/01 – 2019/04 **Targeting Chemotaxis Inhibitory Proteins (CHIP) of Staphylococcus aureus using bio-inspired Iron nanoparticles**
Independent Research Project | Maulana Azad National Institute of Technology | PI: Dr. Khushhali M. Pandey
- Developed iron nanoparticles using plant source against Staphylococcus aureus.
 - Determined the surface topology of the nanoparticles by employing different characterization techniques.
 - Mentored 10 undergraduate students for their bachelor thesis.
 - Addressed the results in a reputed international conference.
- 2018/01 – 2018/05 **Development of Microbial Zombies using synthesised metal nanoparticles**
B.Tech Dissertation | Dr. D.Y. Patil Biotechnology & Bioinformatics Institute | PI: Dr. Swapnil Gaikwad
- Developed and characterized silver nanoparticles to examine their antibacterial activity against *Pseudomonas aeruginosa*.
 - Systematized the protocol for nanoparticle synthesis and reported the observations to a reputed journal

Publications

A readthrough-permissive stop context and C-terminal constraint distinguish AQP4ex from its orthodox aquaporin paralogues

BioRxiv

Sangeet S, Postnova S

Quantitative Modelling of Amyloid- β Dynamics in Brain, CSF, and Plasma During Sleep and Wakefulness

BioRxiv (2026)

Sangeet S, Phillips CL, Hoyos CM, D'Rozario AL, Lucey BP, Postnova S

doi: 10.64898/2026.09.07.749991 [↗](#)

Modelling Amyloid-beta clearance dynamics across sleep-wake cycles in healthy conditions

World Sleep Congress, Sleep Medicine (2026)

Sangeet S, Phillips C, Hoyos C, D'Rozario A, Postnova S

doi: Article 106992 [↗](#)

EVOLVE: A Web Platform for AI-Based Protein Mutation Prediction and Evolutionary Phase Exploration

Journal of Chemical Information and Modelling

Sangeet S, Sinha A, Nair MB, Mahata A, Sarkar R, Roy S

doi: 10.1021/acs.jcim.5c00026 [↗](#)

Bacopa monnieri phytochemicals as promising BACE1 inhibitors for Alzheimer's disease therapy

Scientific Reports

Sangeet S, Khan A

doi: 10.1038/s41598-025-92644-y [↗](#)

Spectroscopic and computational study of molecular interaction of Pexidartinib with homologous mammalian transport proteins

Journal of Molecular Liquids

Ankita Jalan, Amit Kumar Pradhan, **Satyam Sangeet**, N Shaemningwar Moyon

doi: 10.1016/j.molliq.2024.125869 [↗](#)

RNA's Dynamic Conformational Selection and Entropic Allosteric Mechanism in Controlling Cascade Protein Binding Event

Journal of Physical Chemistry Letters (2024)

Amrita Chakraborty, Dibyamanjaree Samant, Raju Sarkar, **Satyam Sangeet**, Sangram Prusty, Susmita Roy

doi: 10.1021/acs.jpcllett.4c00740 [↗](#)

Exploring the interaction of a potent anti-cancer drug Selumetinib with bovine serum albumin: Spectral and computational attributes

Journal of Molecular Recognition (2024)

Jalan A, **Sangeet S**, Pradhan AK, Moyon NS

doi: 10.1002/jmr.3084 [↗](#)

An in-silico approach to identify bioactive phytochemicals from Houttuynia cordata Thunb. As potential inhibitors of human glutathione reductase

Journal of Biomolecular Structure and Dynamics (2023)

Sangeet S, Khan A

doi: 10.1080/07391102.2023.2294181 [↗](#)

Evolution of Sequence and Structure of SARS-CoV-2 Spike Protein: A Dynamic Perspective

ACS Omega (2023)

Sinha A*, **Sangeet S***, Roy S (*equal authorship)

doi: 10.1021/acsomega.3c00944 [↗](#)

The quest for novel drugs against COVID-19: In silico study

Tissue Engineering and Regenerative Medicine, CRC Press

Prajapati S, Zeenat L, **Sangeet S**, Khan A, Pandey KM

doi: 10.1201/9781003536352-18 [↗](#)

In-silico screening of potential phytochemicals against Extracellular Adherence Proteins (EAP) of Staphylococcus aureus from Indian Medicinal Plants

Research Journal of Pharmacy and Technology (2023)

Zeenat L, Prajapati S, **Sangeet S**, Khan A, Pandey KM

doi: 10.52711/0974-360X.2023.00762 [↗](#)

Quantifying Mutational Response to track the evolution of SARS-CoV-2 Spike variants: Introducing a Statistical Mechanics-guided Machine Learning Method

The Journal of Physical Chemistry B (2022)

Sangeet S, Sarkar R, Mohanty SK, Roy S

doi: 10.1021/acs.jpcc.2c04574 [↗](#)

Computational Analysis of Bacopa monnieri (L.) Wettst. compounds for drug development against Neurodegenerative Disorders

Current Computer Aided Drug Design (2022)

Sangeet S, Khan A, Mahanta S, Roy N, Das SK, Mohanta YK, Saravanan M, Tag H, Hui PK

doi: 10.2174/1573409918666221010103652 [↗](#)

Exploratory Data Analysis of Genomic Sequence of Variants of SARS-CoV-2 Reveals Sequence Divergence and Mutational Localisation

Bioinformatics and Biology Insights (2022)

Sangeet S, Khan A

doi: 10.1177/11779322221126294 [↗](#)

Computational Approach to attenuate the virulence of Pseudomonas aeruginosa through Bioinspired silver nanoparticles

3 Biotech, Springer (2022)

Sangeet S, Pawar S, Nawani N, Junarkar M, Gaikwad S

doi: 10.1007/s13205-022-03367-0 [↗](#)

Antibacterial property of Biologically synthesized Iron Nanoparticles against Staphylococcus aureus

International Conference on Advancement in Materials, Manufacturing, and Energy Engineering (ICAMME-2021).

Chapter 58, Springer Nature (2021)

Sangeet S, Khan A, Prabha S, Menaria K

doi:10.1007/978-981-16-8341-1_7 [↗](#)

Drug Development against SARS-CoV-2 from Indian Medicinal Plants - Computational Approach

International Journal of Engineering Applied Sciences and Technology (2020)

Sangeet S, Khan A

doi:10.33564/IJEAST.2020.v05i06.047 [↗](#)

Academic Conference and Co-curricular Activities

2026 **European Sleep Research Society**

Maastricht, Netherlands *Poster Presentation*

2025 **World Sleep Congress**

Singapore *Poster Presentation*

Presented my poster titled "*Modelling Amyloid-beta Dynamics Modulated by Sleep-Wake Cycles*" at the World Sleep Conference held at Singapore

2025 Adelaide	Australasian Chronobiology Society <i>Poster Presentation</i> Delivered an oral presentation for my work titled "Modelling Amyloid-beta Dynamics Modulated by Sleep-Wake Cycles" at the Australasian Chronobiology Society held in Adelaide
2023 Paris	iGEM IISER Kolkata <i>Developing Machine Learning Models for Mutation Prediction in Xanthomonas oryzae</i> Won Gold Medal in iGEM 2023 held in France on the topic "Understanding the Social Language and Virulence Mechanisms of Bacteria: Strategy for Designing Disease Interference for Xanthomonas oryzae"
2018 Pune	Volunteered in Industry-Academia Conclave organised by Dr. D.Y. Patil Biotechnology & Bioinformatics Institute
2017 Pune	Hutington's Disease: Pathophysiology and Genetic causes <i>Poster Presentation</i> Sangeet S, Kapse S, Khan S Dr. D.Y. Patil Biotechnology & Bioinformatics Institute
2016 Pune	Usage of DNA as Memory Storage Device <i>Poster Presentation</i> Sangeet S, Singh D, Kapse S Dr. D.Y. Patil Biotechnology and Bioinformatics Institute

Invited Talks

2023/03	Invited as a Speaker/Tutor to deliver a workshop on "Molecular Docking & Molecular Dynamic Simulations for Drug Discovery" for Faculty Members <i>Part of Faculty Training Program</i> NEMCARE Group of Institutions
2022/01	Invited as a guest lecturer for Science Foundation on the topic "Exploring DNA using Biopython" <i>Deen Dayal Upadhyay College, University of Delhi</i>
2021/02	Workshop on "Molecular Docking in Drug Screening" <i>CompObelisk</i>
2020/08 – 2020/09	Workshop on "Importance of Python & Biopython for Data analysis in Biological Systems" <i>LifeGenBio</i>

Technical Skills

Microbiological techniques	● ● ● ● ●	Analytical Techniques	● ● ● ● ●
Computational and Bioinformatics	● ● ● ● ●	Molecular Biology	● ● ● ● ●
Machine Learning	● ● ● ● ●		

Relevant Courses

Computational Biology

Bioinformatics, Cheminformatics, Drug Screening, Structural Biology, Data Analysis, Computational Neuroscience

Programming

Python, Biopython, R, C-programming, Matlab

Molecular Biology & Nanotechnology

Molecular Biology, Analytical Skills, Immunology, Animal Tissue Culture, Biosensor, and Nanotechnology, Virology

Microbiology and Cell Biology

Microbiology, Mammalian Physiology, Culture techniques, Cell Biology

Bioinformatic Skills

BLAST, Genomic Data Analysis, Database Scrapping, Machine Learning with Omics Data, Protein Sequence Analysis, Sequence Alignments